HASSE–WITT MATRIX VANISHING AND CUBIC SCALAR CANCELLATION ON THE $k = 3$ QUARTIC REDUCTION BRANCH

ARGUS OPEN RESEARCH PROJECT

Abstract. In connection with the comparison of non-ordinary and supersingular reduction primes of abelian varieties, we prove that on the index-7 prime-carry branch at $k = 3$ for the plane quartic family, $k = 3$ forces vanishing of the Hasse–Witt trace condition $H_p = 0$ and exact cancellation of the full cubic scalar concomitant whenever $H_p = 0$.

1. Main Proposition

Proposition 1.1. *For primes $p \equiv 6 \pmod 7$ on the $k = 3$ branch of the quartic family, the Cartier–Manin/Hasse–Witt invariant H_p satisfies the exact polynomial factorization in which the cubic scalar concomitant vanishes identically modulo H_p.*

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